Structural Dynamics of GameStop Equity Pricing: A Multi-Layer Market Microstructure Framework

Abstract

This paper develops a scenario-based market microstructure framework for evaluating price dynamics in GameStop Corp. (NYSE: GME). The framework integrates securities-lending supply constraints, derivative hedging flows, convertible arbitrage positioning, settlement discipline, and warrant-linked capital-structure features into a unified quantitative system. The analysis is not designed to forecast a single deterministic price target. Instead, it evaluates how overlapping structural mechanisms may compress effective float, increase borrow stress, and amplify nonlinear price responses when key thresholds are breached.

The paper contributes four elements. First, it formalizes effective-float compression and borrow-utilization convexity. Second, it develops a reduced-form gamma and liquidity impact model. Third, it provides a worked numerical example and sensitivity analysis. Fourth, it adds an empirical event-study benchmark using the January 2021 GME squeeze episode and a Python appendix implementing both the event study and a Monte Carlo stress simulation.

1 Introduction

Equity price formation in stressed markets is shaped not only by issuer fundamentals but also by the mechanics of trading, clearing, settlement, dealer intermediation, and securities lending. BIS work on market-making and market liquidity emphasizes that dealer balance-sheet constraints and inventory frictions can magnify price moves when liquidity supply becomes thin or one-sided [1, 2]. IOSCO publications on short selling and securities settlement similarly emphasize the importance of settlement discipline, securities borrowing, and safeguards against delivery disruption [4, 3]. In the United States, the SEC's GameStop staff report and Regulation SHO guidance provide a useful institutional benchmark for understanding order handling, options activity, settlement obligations, and the role of failures to deliver

and close-out requirements [5, 6, 7].

GameStop represents an unusual case study because several structural features coexist in a single name:

- a large warrant distribution linked to convertible debt,
- adjusted legacy option deliverables including both shares and warrants,
- concentrated long ownership and directly registered shares,
- meaningful options participation and strike clustering,
- and a borrow market in which supply can tighten discontinuously.

The objective of this paper is to analyze these features as a layered market-structure system rather than as an ordinary valuation problem.

2 Institutional and Regulatory Background

2.1 Market-making, liquidity, and inventory constraints

BIS and CGFS work argues that market-making capacity depends on balance-sheet flexibility, inventory tolerance, and the ability of intermediaries to warehouse risk [1, 2]. When dealers face capacity constraints, price impact becomes more nonlinear. This is directly relevant to a stock in which both directional demand and hedging demand may rise while available float and lendable supply contract.

2.2 Short selling, securities lending, and settlement discipline

IOSCO's short-selling framework highlights two policy concerns that are directly relevant for stressed names: first, the potential for naked short selling to contribute to settlement disruption; second, the need to preserve legitimate securities lending and hedging activity

while maintaining delivery discipline [4]. IOSCO's settlement recommendations likewise note that liquid securities-lending markets can reduce settlement failures, but only when borrowing and delivery disciplines function as intended [3].

In the United States, Regulation SHO Rule 204 imposes close-out requirements for certain failures to deliver, including an extended period for some deemed-owned positions, while the 2009 SEC adopting release emphasized that persistent fails can undermine settlement integrity [6, 7]. These mechanisms matter because borrow stress, recall events, and delivery frictions do not remain purely financing issues once settlement obligations bind.

2.3 GameStop as a microstructure benchmark

The SEC's 2021 staff report on equity and options market structure conditions in early 2021 documented that the GME episode featured extraordinary trading volume, heightened retail participation, elevated options activity, and short covering, but also noted that buying by short-coverers represented only a small fraction of overall buy volume during key phases of the run-up [5]. The same report further stated that staff did not find evidence that a gamma squeeze was the principal driver of the January 2021 episode [5]. This matters for the current framework: the point is not to mechanically relabel any sharp move a gamma squeeze, but to build a system that can separately evaluate float compression, borrow stress, options hedging, and settlement pressure.

3 Structural Framework

3.1 Capital-structure components

The broader structural thesis rests on five interacting components:

1. effective-float contraction through direct registration, recalls, and converted-and-retained shares;

2. borrow-market stress through changes in lendable supply;

3. direct demand shocks from hedge unwinds and covering;

4. reflexive dealer hedging near concentrated strike regions;

5. settlement and delivery discipline once obligations become harder to source.

3.2 Notation

Table 1 standardizes the notation used throughout the paper.

Table 1: Core Variables

Symbol	Definition
S_{out}	Total shares outstanding
F	Public float
DRS	Directly registered shares
L	Total lendable share pool
B	Borrowed shares currently on loan
u	Borrow utilization ratio
W	Total warrants outstanding
C_r	Recalled shares removed from lending supply
α_r	Fraction of recalled shares that were actually lendable
C_f	Converted-and-retained shares from financed flips
E_{lock}	Exercised or otherwise locked shares removed from effective float
H_u	Hedge-unwind purchase demand
H_g	Dealer hedge demand generated by gamma exposure
FTD_t	Delivery-failure stress term at time t
NBP_t	Net buy pressure at time t
P_t	Stock price at time t
K_w	Warrant strike price
ADV	Average daily volume

4 Quantitative Modeling Framework

4.1 Effective float

The first-order effective float is defined as

$$F_{\text{eff}} = F - DRS - C_r - C_f - E_{\text{lock}}. \tag{1}$$

Equation (1) converts reported float into the subset of shares realistically available to absorb marginal net demand. This is not a legal-float measure; it is a stress-liquidity measure.

4.2 Float compression ratio

The float compression ratio is

$$FCR = \frac{F_{\text{eff}}}{F}. \tag{2}$$

The locked-float ratio is

$$LFR = 1 - FCR. \tag{3}$$

As FCR falls, the same amount of incremental demand produces larger price impact.

4.3 Borrow utilization and recall shock

Baseline borrow utilization is

$$u = \frac{B}{L}. \tag{4}$$

A recall shock modifies the lendable pool and therefore utilization:

$$u' = \frac{B}{L - \alpha_r C_r}. \tag{5}$$

Equation (5) captures why recall events can matter even if reported short interest does not change. The numerator remains fixed initially while the denominator contracts.

4.4 Borrow-fee convexity

A stylized reduced-form borrow-fee function is

$$r_b = \alpha_0 + \alpha_1 u + \alpha_2 u^2 + \varepsilon, \tag{6}$$

where $\alpha_2 > 0$ captures convexity as utilization rises toward stressed levels.

4.5 Gamma exposure

For option line i, share-equivalent gamma exposure can be approximated as

$$GEX_i(P) = \Gamma_i(P)\, OI_i\, M\, s_i, \tag{7}$$

where OI_i is open interest, M is the contract multiplier, and $s_i \in \{-1, +1\}$ encodes the sign convention for dealer positioning.

Aggregate gamma exposure is

$$GEX_{\text{total}}(P) = \sum_i GEX_i(P). \tag{8}$$

A local hedge-demand approximation is

$$\frac{dH_g}{dP} \approx GEX_{\text{total}}(P). \tag{9}$$

4.6 Reduced-form price impact

The central reduced-form return specification is

$$dP_t = \mu\, dt + \sigma\, dW_t + \theta_1 \left(\frac{NBP_t}{F_{\text{eff},t}} \right) + \theta_2 \max(0, u_t - u^\star) + \theta_3 GEX_t + \theta_4 FTD_t, \tag{10}$$

where $u^\star$ is the utilization threshold above which borrow stress becomes convex.

Equation (10) does not claim structural invariance. Its purpose is to make the framework explicit enough to simulate.

4.7 Net buy pressure

For a given simulation step, net buy pressure is

$$NBP_t = H_{u,t} + H_{c,t} + H_{g,t} + H_{s,t}, \tag{11}$$

where $H_{c,t}$ is covering demand and $H_{s,t}$ is settlement-stress demand.

5 Worked Numerical Example

This section shows one internally consistent scenario using the same framework.

5.1 Scenario inputs

Assume the following:

$$
\begin{aligned}
F &= 409 \text{ million}, \\
DRS &= 67 \text{ million}, \\
C_r &= 20 \text{ million}, \\
C_f &= 35 \text{ million}, \\
E_{\text{lock}} &= 3.5 \text{ million}, \\
L &= 105 \text{ million}, \\
B &= 68 \text{ million}, \\
\alpha_r &= 0.80, \\
H_u &= 35 \text{ million}, \\
H_c &= 10 \text{ million}, \\
H_g &= 8 \text{ million}, \\
H_s &= 4 \text{ million}.
\end{aligned}
$$

5.2 Effective float and compression

Using equation (1),

$$F_{\text{eff}} = 409 - 67 - 20 - 35 - 3.5 = 283.5 \text{ million}. \tag{12}$$

Using equations (2) and (3),

$$FCR = \frac{283.5}{409} = 0.693, \qquad LFR = 0.307. \tag{13}$$

5.3 Borrow utilization after recall

Using equation (5),

$$u' = \frac{68}{105 - (0.80 \times 20)} = \frac{68}{89} = 0.764. \tag{14}$$

Thus, utilization rises to 76.4%.

5.4 Net buy pressure

Using equation (11),

$$NBP = 35 + 10 + 8 + 4 = 57 \text{ million}. \tag{15}$$

Relative demand is therefore

$$\frac{NBP}{F_{\text{eff}}} = \frac{57}{283.5} = 0.201. \tag{16}$$

If baseline average daily volume is 7 million shares, then the volume multiple is

$$\frac{57}{7} = 8.14 \times ADV. \tag{17}$$

5.5 Stylized price band

If a liquidity multiplier $\Lambda \in [1.2, 1.8]$ is applied to the relative-demand term in equation (16), the implied proportional move range is

$$0.201 \times 1.2 = 24.1\%, \qquad 0.201 \times 1.8 = 36.2\%. \tag{18}$$

From a \$24 baseline:

$$24 \times 1.241 = 29.78, \qquad 24 \times 1.362 = 32.69. \tag{19}$$

Table 2: Worked Numerical Example

Metric	Formula	Result
Effective float	Eq. (12)	283.5M
Float compression ratio	Eq. (13)	0.693
Locked-float ratio	Eq. (13)	0.307
Post-recall utilization	Eq. (14)	76.4%
Net buy pressure	Eq. (15)	57M
Net buy pressure / effective float	Eq. (16)	20.1%
Volume multiple	Eq. (17)	8.14x
Stylized price band from \$24	Eqs. (18)–(19)	\$29.78–\$32.69

6 Sensitivity Analysis

Table 3 varies the four most important parameters in the framework: recall volume, converted-and-retained shares, hedge-unwind demand, and gamma-driven dealer hedge demand.

Table 3: Low, Base, and High Sensitivity Cases

Case	C_r	C_f	H_u	H_g	F_{eff}	Price Range
Low	10M	20M	15M	3M	308.5M	\$26.99–\$28.48
Base	20M	35M	35M	8M	283.5M	\$29.78–\$32.69
High	30M	50M	55M	15M	258.5M	\$33.36–\$38.04

The low-case utilization arithmetic is

$$u'_{\text{low}} = \frac{68}{105 - (0.80 \times 10)} = \frac{68}{97} = 0.701, \tag{20}$$

or 70.1%.

The high-case utilization arithmetic is

$$u'_{\text{high}} = \frac{68}{105 - (0.80 \times 30)} = \frac{68}{81} = 0.840, \tag{21}$$

or 84.0%.

Table 4: One-at-a-Time Sensitivity Shocks

Variable Shock	Adjusted Value	$\frac{NBP}{F_{\text{eff}}}$	Stylized Price Range
Lower recall	$C_r = 10$M	19.4%	\$29.58–\$32.38
Higher recall	$C_r = 30$M	20.8%	\$29.99–\$32.99
Lower conversion lock	$C_f = 20$M	19.1%	\$29.50–\$32.25
Higher conversion lock	$C_f = 50$M	21.2%	\$30.10–\$33.17
Lower hedge unwind	$H_u = 15$M	13.1%	\$27.76–\$29.67
Higher hedge unwind	$H_u = 55$M	27.2%	\$31.83–\$35.76
Lower gamma demand	$H_g = 3$M	18.3%	\$29.28–\$31.90
Higher gamma demand	$H_g = 15$M	22.6%	\$30.51–\$33.76

7 Empirical Backtest: January 2021 GME Squeeze Event

7.1 Purpose

This section does not claim that the January 2021 event is a perfect analog for the current setup. The SEC's report makes clear that the 2021 run-up reflected a complex mix of retail participation, elevated volume, options activity, and short covering, while staff did not find evidence that gamma squeeze mechanics were the primary driver [5]. The historical event is used here as a benchmark stress episode for validating regime classification.

7.2 Event-study design

Define daily log returns for asset i as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \tag{22}$$

Define benchmark-adjusted abnormal returns relative to SPY as

$$AR_t = r_{GME,t} - r_{SPY,t}. \tag{23}$$

Cumulative abnormal return over event window $[t_0, t_1]$ is

$$CAR_{[t_0,t_1]} = \sum_{t=t_0}^{t_1} AR_t. \tag{24}$$

For a simple price-ratio benchmark, cumulative raw return is

$$R_{[t_0,t_1]} = \frac{P_{t_1}}{P_{t_0}} - 1. \tag{25}$$

7.3 Illustrative benchmark levels

Using widely cited daily historical price series, GME closed at \$7.86 on January 13, 2021, \$16.21 on January 22, 2021, \$86.88 on January 27, 2021, and \$81.25 on January 29, 2021 [8]. Using equation (25), the raw return from January 13 to January 27 is approximately

$$R_{[\text{2021-01-13, 2021-01-27}]} = \frac{86.88}{7.86} - 1 = 10.053, \tag{26}$$

or approximately 1005.3%.

Table 5: Illustrative January 2021 Event Benchmarks

Date	Close	Reference Return vs. Jan. 13
2021-01-13	\$7.86	0.0%
2021-01-22	\$16.21	106.2%
2021-01-27	\$86.88	1005.3%
2021-01-29	\$81.25	933.7%

7.4 Backtest interpretation

The event-study benchmark is used here in three ways:

1. to calibrate regime labels in the simulation;

2. to compare stylized model outputs with a known stress episode;

3. to distinguish ordinary compression from true disorderly squeeze dynamics.

A practical regime mapping is:

- **Regime 0: Dormant** — no threshold breach and low abnormal returns,
- **Regime 1: Compression** — float tightens, utilization rises, but event-style abnormal returns remain moderate,
- **Regime 2: Reflexive** — price breaches key levels and hedging/covering feedback becomes self-reinforcing,
- **Regime 3: Disorderly Squeeze** — event-study abnormal returns and volume resemble the January 2021 stress episode.

8 Monte Carlo Squeeze Probability Model

8.1 State variables

The simulation draws from distributions over:

$$
\begin{aligned}
C_r &\sim \text{Discrete/Scenario}, \\
\alpha_r &\sim \text{Uniform}(0.60, 0.95), \\
\phi_f &\sim \text{Discrete/Scenario}, \\
\phi_w &\sim \text{Conditional Scenario}, \\
S_{\text{hidden}} &\sim \text{Scenario Layer}, \\
Q_{\text{liq}} &\sim \text{Lognormal}, \\
GEX_t &\sim \text{State-dependent function of } P_t.
\end{aligned}
$$

8.2 Simulation sequence

For each path:

1. draw structural inputs;
2. compute F_{eff} using equation (1);
3. compute post-shock utilization using equation (5);
4. estimate first-wave buy pressure using equation (11);
5. update price using equation (10);
6. if $P_t \geq K_w$, allow second-wave warrant and gamma responses to activate;
7. iterate through the horizon and classify the path by regime.

8.3 Outputs

The principal outputs are:

- probability of entering Compression, Reflexive, or Disorderly Squeeze regimes,
- price-path distributions over 20-, 40-, and 60-day windows,
- sensitivity of results to recall size, float lock-up, and gamma amplification.

9 Conclusion

The framework presented here does not assert that any single mechanism alone is sufficient to produce extreme price dislocation. Instead, it shows how float compression, borrow stress, direct buy pressure, gamma-sensitive dealer hedging, and settlement discipline may interact in a path-dependent way.

This matters because market stress is often nonlinear. When the effective float shrinks, borrow utilization rises, and net demand is forced through a constrained supply base, the market can transition from ordinary liquidity dynamics into reflexive repricing. The January 2021 GME episode demonstrates the scale of abnormal returns that can emerge during such stress, even if the precise driver mix differs from later or future events [5, 8].

A Python Appendix: Event Study and Monte Carlo Simulation

The following Python script downloads GME and SPY data, computes an event study around the January 2021 squeeze episode, and runs a stylized Monte Carlo simulation based on the structural model in the main text.

Listing 1: Python event-study and Monte Carlo appendix

```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt


# ------------------------------
# 1. Download historical data
# ------------------------------
start_date = "2020-12-01"
end_date = "2021-02-10"


tickers = ["GME", "SPY"]
data = yf.download(tickers, start=start_date, end=end_date, auto_adjust=False)["
    Adj Close"]
data = data.dropna()
```

```
# ----------------------------
# 2. Event-study calculations
# ----------------------------
returns = np.log(data / data.shift(1)).dropna()
returns["AR"] = returns["GME"] - returns["SPY"]

event_start = "2021-01-13"
event_end = "2021-01-29"

event_window = returns.loc[event_start:event_end].copy()
event_window["CAR"] = event_window["AR"].cumsum()

summary = pd.DataFrame({
    "Close_GME": data["GME"].loc[[event_start, "2021-01-22", "2021-01-27",
        event_end]],
})
summary["Raw_Return_vs_Event_Start"] = summary["Close_GME"] / summary["Close_GME"
    ].iloc[0] - 1.0

print("\nEvent-study benchmark table:")
print(summary)

print("\nCumulative abnormal return over event window:")
print(event_window["AR"].sum())

# ----------------------------
# 3. Structural parameter setup
# ----------------------------
```

```
BASE_FLOAT = 409_000_000
BASE_DRS = 67_000_000
BASE_LENDABLE = 105_000_000
BASE_BORROWED = 68_000_000
BASE_ELOCK = 3_500_000
BASE_PRICE = 24.0
WARRANT_STRIKE = 32.0
ADV = 7_000_000

# ----------------------------
# 4. Monte Carlo configuration
# ----------------------------
n_sims = 10000
horizon = 40
dt = 1.0

rng = np.random.default_rng(42)

def draw_recall():
    return rng.choice([0, 10_000_000, 20_000_000, 30_000_000], p=[0.20, 0.30,
        0.30, 0.20])

def draw_alpha_r():
    return rng.uniform(0.60, 0.95)

def draw_converted_retained():
    return rng.choice([10_000_000, 20_000_000, 35_000_000, 50_000_000], p=[0.20,
        0.30, 0.30, 0.20])
```

```
def draw_hidden_short_layer():
    # proportion of float
    return rng.choice([0.00, 0.20, 0.40, 0.60], p=[0.30, 0.30, 0.25, 0.15])

def draw_liquidity_multiplier():
    # reduced-form impact coefficient
    return rng.uniform(1.2, 1.8)

def gamma_term(price):
    # stylized gamma response around key strike regions
    if price < 28:
        return 0.0
    elif price < 32:
        return 0.002 * (price - 28)
    elif price < 40:
        return 0.008 + 0.004 * (price - 32)
    else:
        return 0.040

results = []

for sim in range(n_sims):
    price = BASE_PRICE
    recall = draw_recall()
    alpha_r = draw_alpha_r()
    converted_retained = draw_converted_retained()
    hidden_short_layer = draw_hidden_short_layer()
    liquidity_mult = draw_liquidity_multiplier()
```

```
# effective float
effective_float = BASE_FLOAT - BASE_DRS - recall - converted_retained -
    BASE_ELOCK
effective_float = max(effective_float, 50_000_000)

# utilization after recall
adjusted_lendable = BASE_LENDABLE - alpha_r * recall
adjusted_lendable = max(adjusted_lendable, 1_000_000)
utilization = BASE_BORROWED / adjusted_lendable

path = [price]

for t in range(horizon):
    # first-wave demand
    hedge_unwind = rng.choice([15_000_000, 35_000_000, 55_000_000], p=[0.30,
        0.50, 0.20]) / horizon
    covering = rng.choice([4_000_000, 10_000_000, 18_000_000], p=[0.30, 0.50,
        0.20]) / horizon

    # second-wave activation if threshold crossed
    if price >= WARRANT_STRIKE:
        warrant_demand = rng.choice([3_000_000, 6_000_000, 10_000_000], p
            =[0.35, 0.45, 0.20]) / horizon
        settlement_stress = rng.choice([2_000_000, 4_000_000, 8_000_000], p
            =[0.35, 0.45, 0.20]) / horizon
    else:
        warrant_demand = 0.0
        settlement_stress = rng.choice([0, 1_000_000, 2_000_000], p=[0.50,
            0.30, 0.20]) / horizon
```

```
    gex = gamma_term(price)
    gamma_demand = gex * effective_float / horizon

    net_buy_pressure = hedge_unwind + covering + warrant_demand +
        settlement_stress + gamma_demand

    # reduced-form drift/shock
    noise = rng.normal(0.0, 0.03)
    borrow_stress = max(0.0, utilization - 0.75)

    dprice = (
        noise
        + liquidity_mult * (net_buy_pressure / effective_float)
        + 0.10 * borrow_stress
        + 0.05 * hidden_short_layer
    )

    price = max(0.5, price * (1.0 + dprice))
    path.append(price)

final_price = path[-1]
peak_price = max(path)

# regime classification
if peak_price < 28:
    regime = "Dormant"
elif peak_price < 32:
    regime = "Compression"
```

```
    elif peak_price < 60:
        regime = "Reflexive"
    else:
        regime = "Disorderly Squeeze"

    results.append({
        "final_price": final_price,
        "peak_price": peak_price,
        "effective_float": effective_float,
        "utilization": utilization,
        "regime": regime,
        "recall": recall,
        "converted_retained": converted_retained,
        "hidden_short_layer": hidden_short_layer
    })

results = pd.DataFrame(results)

print("\nMonte Carlo regime probabilities:")
print(results["regime"].value_counts(normalize=True).sort_index())

print("\nMonte Carlo descriptive statistics:")
print(results[["final_price", "peak_price", "effective_float", "utilization"]].
    describe())

# -----------------------------
# 5. Save outputs
# -----------------------------
results.to_csv("gme_monte_carlo_results.csv", index=False)
```

```
event_window.to_csv("gme_event_window_returns.csv")


# ----------------------------
# 6. Plot event-study CAR
# ----------------------------
plt.figure(figsize=(9, 5))
plt.plot(event_window.index, event_window["CAR"], linewidth=2)
plt.title("GME Event Study: Cumulative Abnormal Return vs SPY")
plt.xlabel("Date")
plt.ylabel("CAR")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("gme_event_study_car.png", dpi=200)


# ----------------------------
# 7. Plot Monte Carlo histogram
# ----------------------------
plt.figure(figsize=(9, 5))
plt.hist(results["peak_price"], bins=60)
plt.title("Monte Carlo Distribution of Peak Price")
plt.xlabel("Peak Price")
plt.ylabel("Frequency")
plt.tight_layout()
plt.savefig("gme_peak_price_histogram.png", dpi=200)
```

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